

General Relativity Coursework

Section (a): Numerical Calculation of Particle Orbits in the Schwarzschild Spacetime

Original Script/Code (with annotations)

```
[2]: import numpy as np
import matplotlib.pyplot as plt

# Parameters
R0 = 7 # Initial Radius, has to be greater than or equal to 6
alpha = 12.25 # Angular Momentum, has to be greater than 12, for circular orbit, has to equal R0/(1-3/R0)
dRdTau0 = 0 # Boundary Condition (No radial Motion at Tau=0)
M = 100 # Mass

def d2RdTau2(R): # Defines the derivative as a function
    return -1/((R**2)*(M**2)) + alpha/((R**3)*(M**2)) - 3*alpha/((R**4)*(M**2))

endTau = 1000 # End Proper Time
steps = 2000 # Steps for controllable degree of accuracy
taus = np.linspace(0, endTau, steps) # An array with size of steps with equally spaced numbers between 0 and the end tau
h = taus[1] - taus[0] # Step Length

R = np.zeros(len(taus)) # Creates an array of 0's with size of how many taus there are
dRdTau = np.zeros(len(taus)) # Same as Last Line but for radial motion
R[0] = R0 # Sets initial R
dRdTau[0] = dRdTau0 # Sets initial radial motion

for i in range(len(taus)-1): # Start of Runge-Kutta 4
    k1_R = dRdTau[i] # Setting initial radial motion
    k1_dRdTau = d2RdTau2(R[i]) # Setting initial radial acceleration
    k2_R = dRdTau[i] + 0.5 * h * k1_dRdTau # Estimated radial motion halfway to next step
    k2_dRdTau = d2RdTau2(R[i] + 0.5 * h * k1_R) # Estimated radial acceleration halfway to next step
    k3_R = dRdTau[i] + 0.5 * h * k2_dRdTau # Use previous estimate to make another
    k3_dRdTau = d2RdTau2(R[i] + 0.5 * h * k2_R) # Use previous estimate to make another
    k4_R = dRdTau[i] + h * k3_dRdTau # Estimated radial motion for end of step
    k4_dRdTau = d2RdTau2(R[i] + h * k3_R) # Estimated radial acceleration for end of step
    R[i+1] = R[i] + (h / 6.0) * (k1_R + 2*k2_R + 2*k3_R + k4_R) # By combining all estimates we get a weighted average for R after the step
    dRdTau[i+1] = dRdTau[i] + (h / 6.0) * (k1_dRdTau + 2*k2_dRdTau + 2*k3_dRdTau + k4_dRdTau) # By combining all estimates we get a weighted average for radial motion after the step

# Next section I added for part 6 only:
x = np.zeros(len(taus)) # Makes an array of 0's as with R
y = np.zeros(len(taus)) # Makes an array of 0's as with R
for i in range(len(taus)):
    x[i] = R[i]*np.cos(0.1*taus[i]) # Uses given theta to get x
    y[i] = R[i]*np.sin(0.1*taus[i]) # Uses given theta to get y

# (inactive due to part 6) plt.plot(taus, R, label=f"$R(\tau)$", $\alpha$={alpha}$, $R_0$={R0}$, $M$={M}$") # Plots R against tau with label for graph and parameter values
# plt.xlabel(r"Proper time $\tau$") # Labels tau axis
# plt.ylabel(r"Radius $R$") # Labels R axis

plt.plot(x, y, label=f"$x(\tau)$, $y(\tau)$", $\alpha$={alpha}$, $R_0$={R0}$, $M$={M}$") # Plots parameterised x by y and labels parameters
plt.xlabel(r"x") # Labels x axis
plt.ylabel(r"y") # Labels y axis

plt.axhline(y=0, color='k') # Making axis clearer
plt.axvline(x=0, color='k') # Making axis clearer

plt.title("Radial Motion of a Particle Orbiting a Black Hole in Schwarzschild Spacetime") # Title describing what we're seeing
plt.grid(True) # Adds a grid
plt.legend(loc='upper center', bbox_to_anchor=(0.5, -0.15), fontsize=14) # Moving legend outside of graph and making it bigger
plt.tight_layout() # Makes the graph fit nicer
figNum="Figure 16" # Figure number to label the graph
plt.figtext(0.5, 0.01, figNum, wrap=True, horizontalalignment='center', fontsize=14) # Adds the label with figure number
plt.show() # Shows graph
```

Some Representative Plots of Results and Explanations

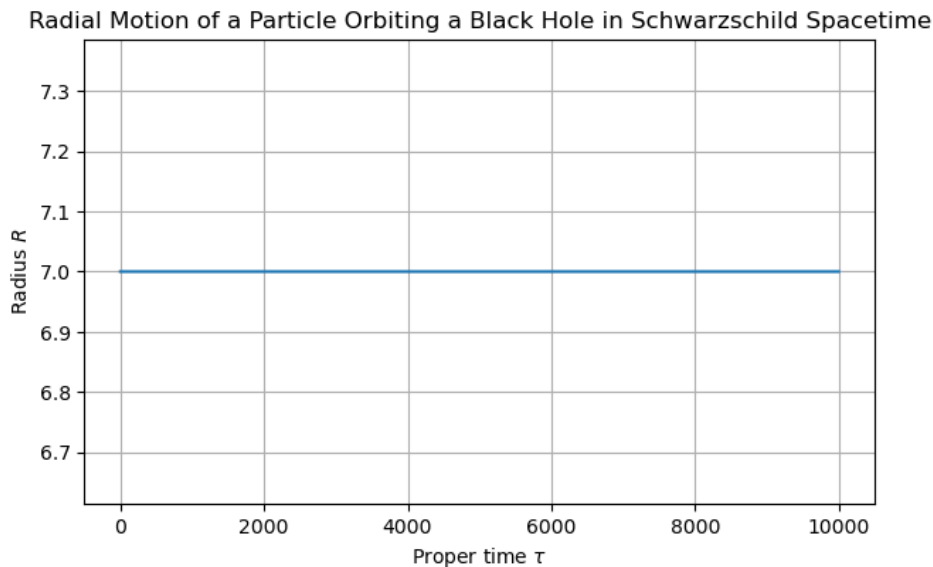


Figure 1

In figure 1, I chose a sensible initial

R (above 6) and set $\alpha = \frac{R_0}{1 - \frac{3}{R_0}}$ to

give a perfectly circular orbit that is not affected by mass and keeps a constant R (the initial R).

I have added an additional graph (figure 2) the exact same but with $M=1000$ instead to show that even with a drastic change in mass, the orbit will remain the same in this case.

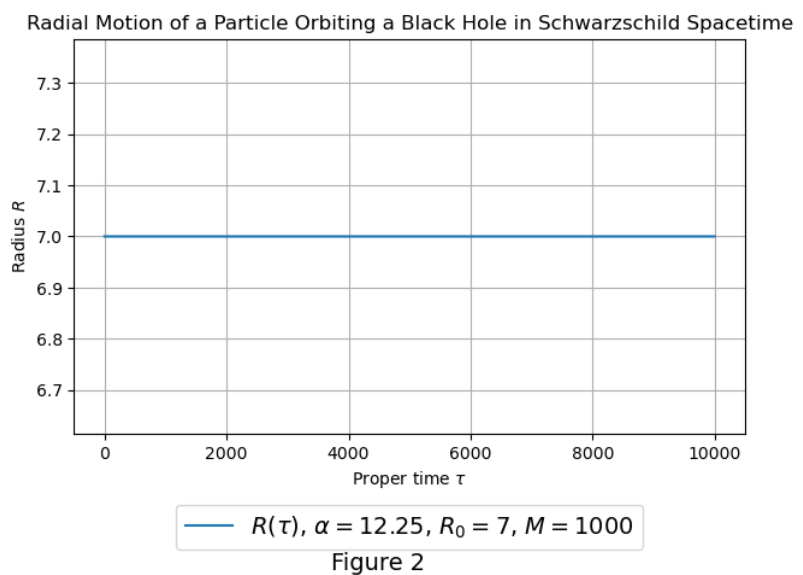


Figure 2

Radial Motion of a Particle Orbiting a Black Hole in Schwarzschild Spacetime

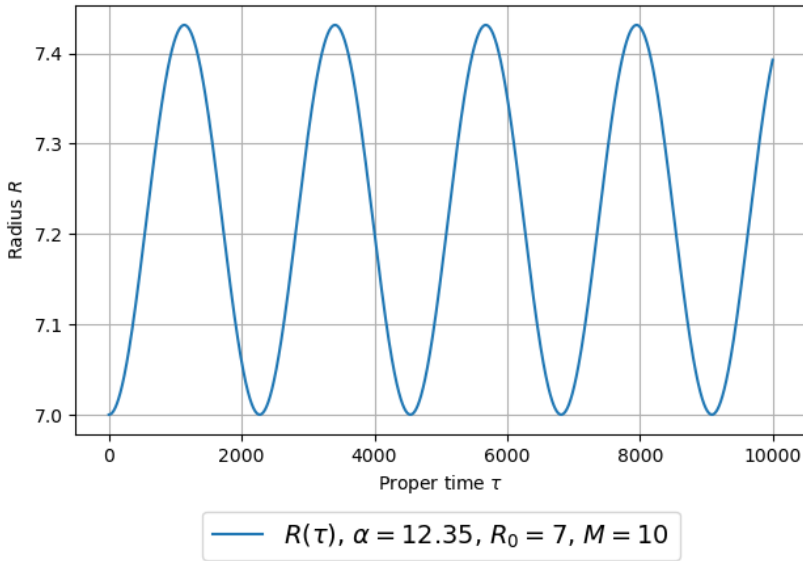


Figure 3

In figure 3, I kept the same initial R , however changed

$$\alpha \text{ to be } 12 < \alpha < \frac{R_0}{1 - \frac{3}{R_0}}.$$

This result shows that the radius does change, however it is periodic, showing that it is orbiting the black hole but in an elliptical orbit instead of a perfectly circular one.

Maximum R is also the initial value in this plot, which is of note because we will see that the opposite is true when α is changed to be bigger than the equation involving initial R . Also note the distance between minimum and maximum R as this is also something I will mention and test later relating to the value of α .

In figure 4, I changed α to be the opposite of the last

$$\text{plot, } 12 < \frac{R_0}{1 - \frac{3}{R_0}} < \alpha. \text{ As}$$

before, the orbit is elliptical, however this time the initial R is the minimum rather than the maximum. You will notice that the distance between the minimum and maximum R is the same as the last plot and there is a

reason for this, the reason being that the difference between α and $\frac{R_0}{1 - \frac{3}{R_0}}$ is the same (0.1) and in later plots I will show the effect of making this difference larger or smaller.

Radial Motion of a Particle Orbiting a Black Hole in Schwarzschild Spacetime

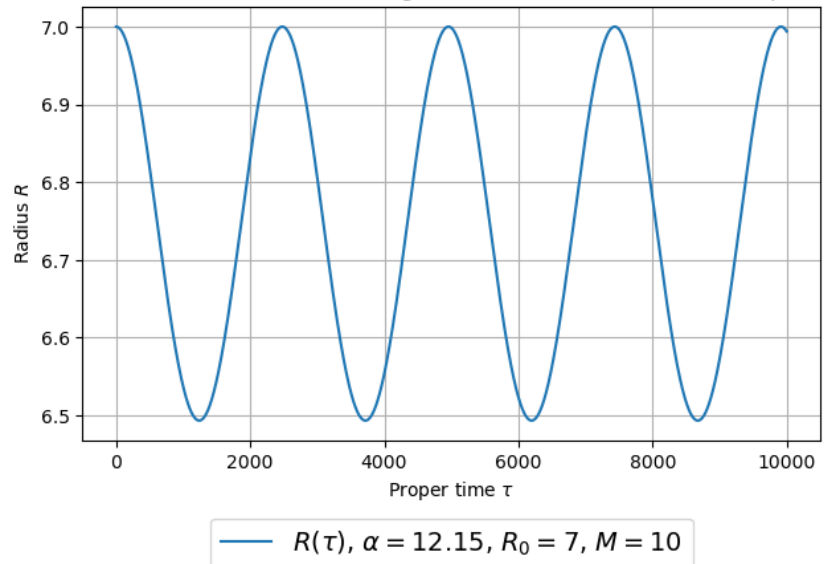
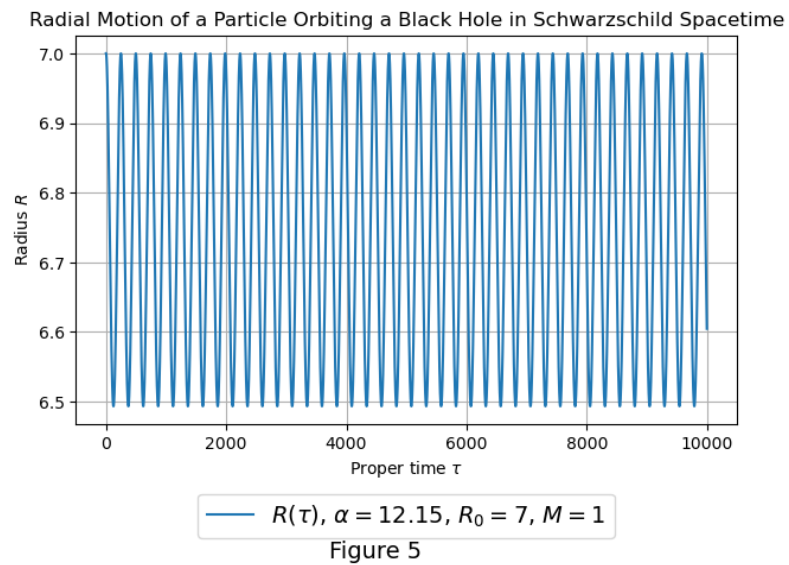
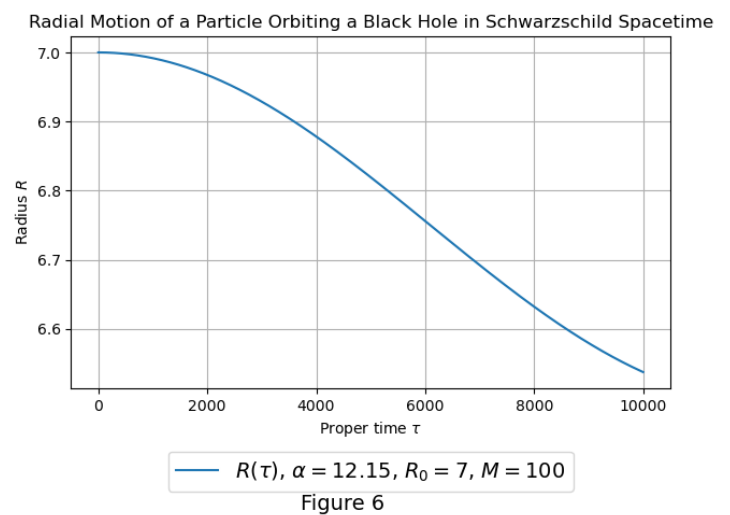


Figure 4

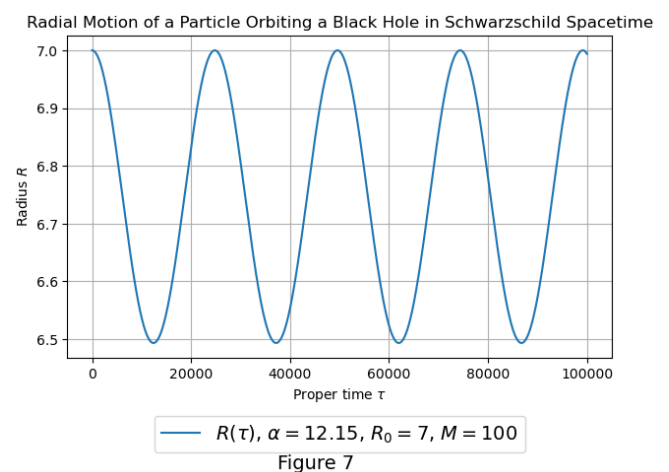
Now using the previous plot, I will change M and show the effect M has on the period of the orbit, keeping the proper time axis the same length and making M smaller, say $M=1$, gives figure 5.



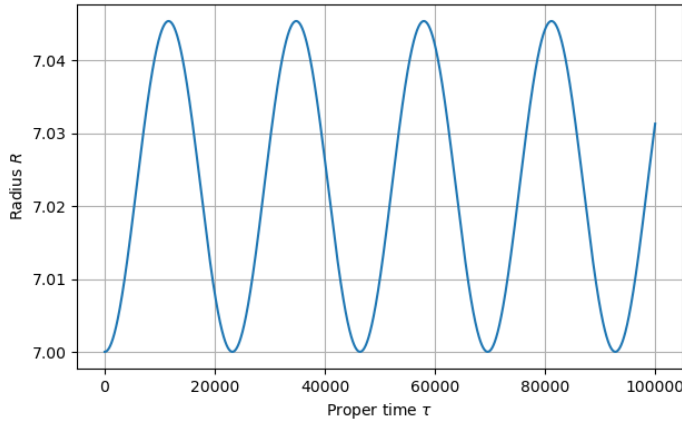
Decreasing M decreases the period of orbit, so you'd imagine increasing M would increase the period of orbit, now setting $M=100$ and keeping the proper time axis the same length gives figure 6.



As you can see the period of orbit has become so long that it won't fit on the current axis for proper time, so in figure 7 I will extend it to show it is still orbiting.



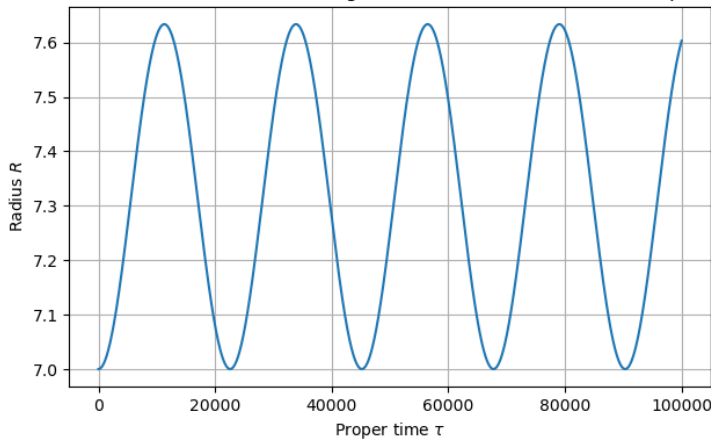
Radial Motion of a Particle Orbiting a Black Hole in Schwarzschild Spacetime



$$R(\tau), \alpha = 12.26, R_0 = 7, M = 100$$

Figure 8

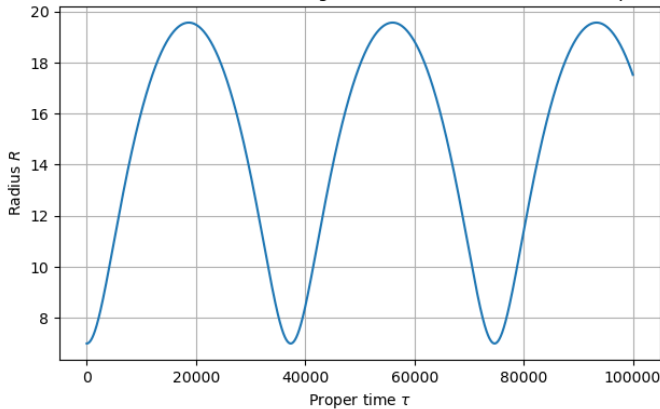
Radial Motion of a Particle Orbiting a Black Hole in Schwarzschild Spacetime



$$R(\tau), \alpha = 12.4, R_0 = 7, M = 100$$

Figure 9

Radial Motion of a Particle Orbiting a Black Hole in Schwarzschild Spacetime



$$R(\tau), \alpha = 15, R_0 = 7, M = 100$$

Figure 10

Now I will show in figures 8, 9 and

10 that α 's distance from $\frac{R_0}{1 - \frac{3}{R_0}}$

affects the difference between maximum and minimum R .

As can be seen in figure 8, the difference between minimum and maximum R is roughly 0.045

when α is only 0.01 higher than

$\frac{R_0}{1 - \frac{3}{R_0}}$. And in figure 9 the

difference between minimum and maximum R is roughly 0.65

when α is 0.15 higher than

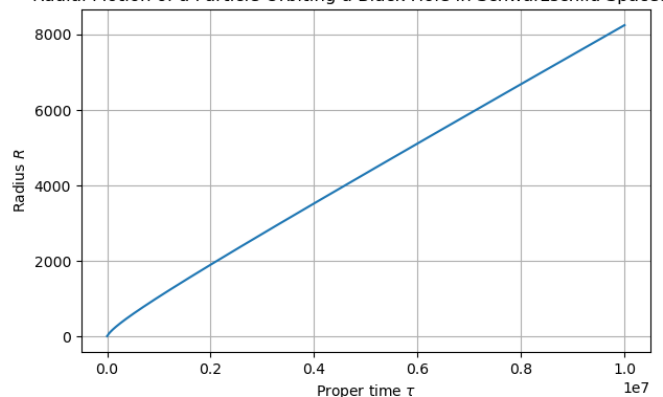
$\frac{R_0}{1 - \frac{3}{R_0}}$. However, if it is increased

past a certain limit, the particle will not orbit and fly away into space as can be seen in figure 11.

This makes sense as α is the

angular momentum, so if higher enough would not be pulled into orbit by the black hole.

Radial Motion of a Particle Orbiting a Black Hole in Schwarzschild Spacetime



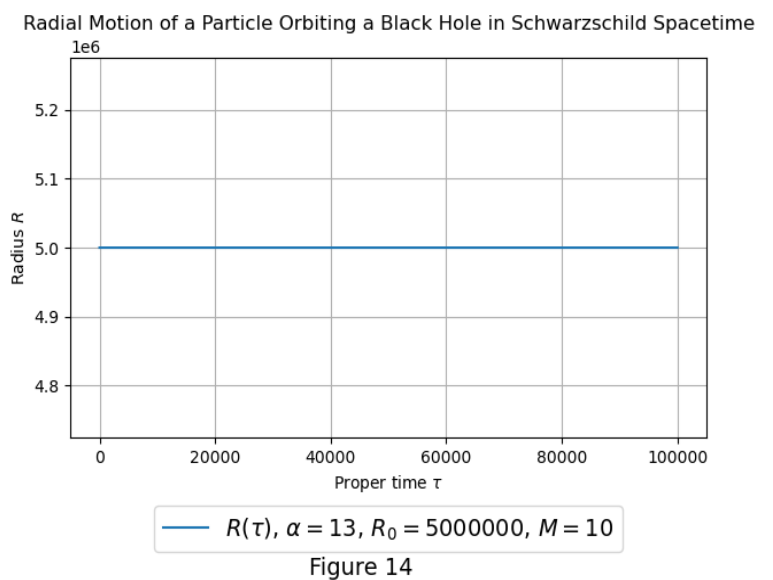
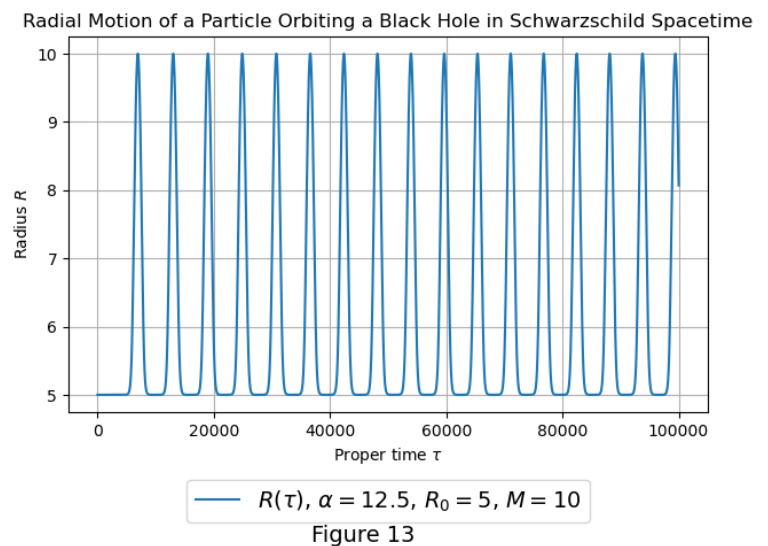
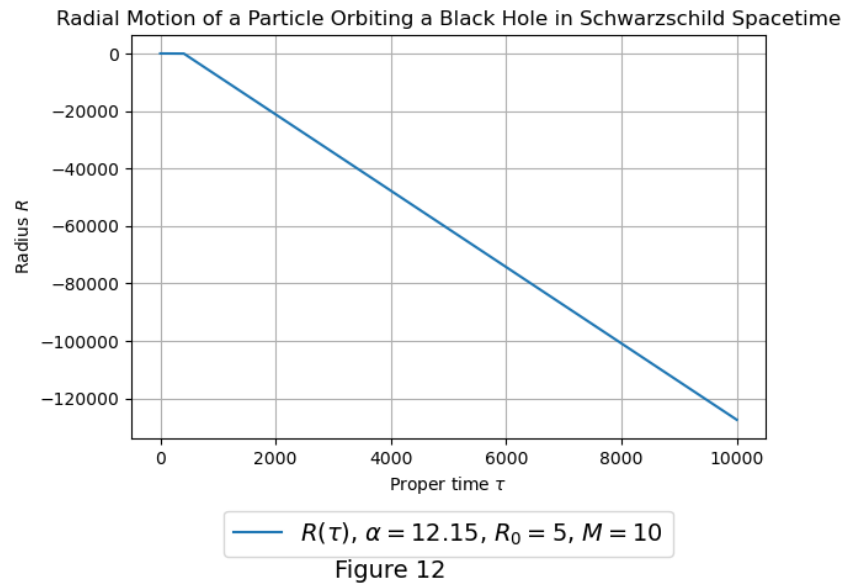
$$R(\tau), \alpha = 20, R_0 = 7, M = 100$$

Figure 11

In figure 12, I set initial $R < 6$, which, according to Schutz, would not be stable and this can be seen in the figure as the particle falls into the black hole and the equation breaks down.

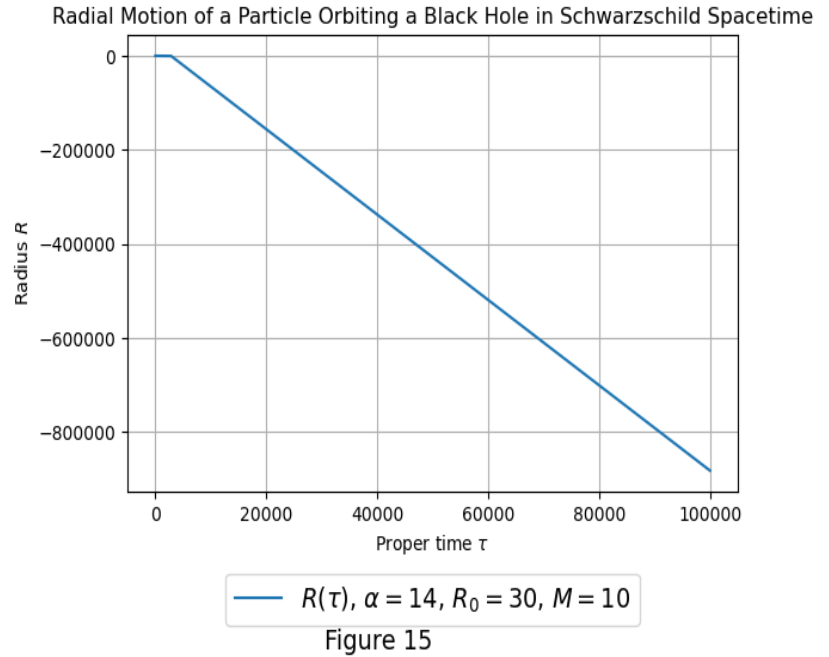
However, when α is changed accordingly to the equation for circular orbit, it gives the result in figure 13, which is a very interesting orbit that seems to keep the same R for a little while before moving far away, then coming back. This does fit with what I have read in Schutz since it only calls $6M$ the minimum for a stable circular orbit and never says it can't have a stable orbit at all like I initially misread.

However, interestingly if I set initial R to be much larger than all my current figures in figure 14, the equation will keep the same R over proper time, but in this case, I believe it is not an orbit but instead the particle remaining static due to no initial movement and being far enough from the black hole to not be affected by its bending of spacetime. This remains the same no matter what α or M , α making sense but M being a little confusing as if the



mass becomes big enough you would expect it to eventually reach the large initial R , but this doesn't happen from my tests, even for absurdly high M 's relative to initial R .

When R is large but small enough to remain under the black hole's effects, if α is too small, the particle will fall into the black hole as seen in figure 15, like figure 12 with R smaller than 6.



Explanation of Results

Overall, based on my testing with my code, my annotations of the figures I provided and reading Schutz I have come to these conclusions about the radial motion of a particle orbiting a black hole in Schwarzschild Spacetime:

Based on initial R , I can observe that if $\alpha = \frac{R_0}{1 - \frac{3}{R_0}}$ with $R_0 > 6$, the orbit is circular, and R does not change over proper time.

If $12 < \alpha < \frac{R_0}{1 - \frac{3}{R_0}}$ with $R_0 > 6$, the orbit is elliptical with R_0 being the maximum R .

If $12 < \frac{R_0}{1 - \frac{3}{R_0}} < \alpha$ with $R_0 > 6$, the orbit is elliptical with R_0 being the minimum R .

Changing M affects the period of the orbit, larger M increases the period and smaller M decreases the period.

Making initial R very large will lead to it maintaining the same R over proper time but instead of being because it's in a circular orbit it's because it has no initial motion and isn't affected by the black hole from that far away.

If initial $R < 6$, then it can still hold an orbit, just not a stable circular one. This orbit is very interesting and has R stay consistent at some parts before returning to a somewhat elliptical orbit.

Adding Angular Position and Plotting in Cartesian Coordinates

Upon considering the angular position θ and setting it equal to $0.1 \cdot \tau$, then parameterising the coordinates into cartesian coordinates and plotting them, the result was the plot in figure 16, which is a stable circular orbit like the one in figure 11.4 in Schutz.

Upon changing α to be either above or below the equation for stable circular orbit, the plot in figure 17 shows that the orbit is elliptical but the ellipse rotates after each period, meaning the plot over enough time will be a thick perfect circle with the thickness being the difference between the initial R and maximum/minimum R controlled by how close α is to the equation for a circular orbit, as seen in figures 8 and 9. As far as I have tested I haven't yet found an α with period that doesn't cover all angles to make the perfect

Radial Motion of a Particle Orbiting a Black Hole in Schwarzschild Spacetime

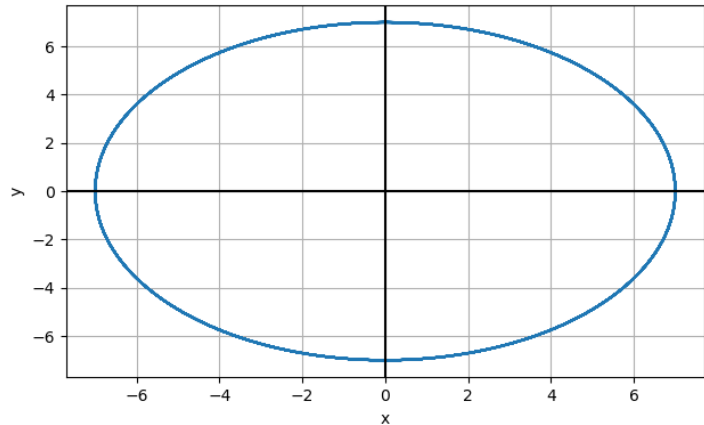


Figure 16

Radial Motion of a Particle Orbiting a Black Hole in Schwarzschild Spacetime

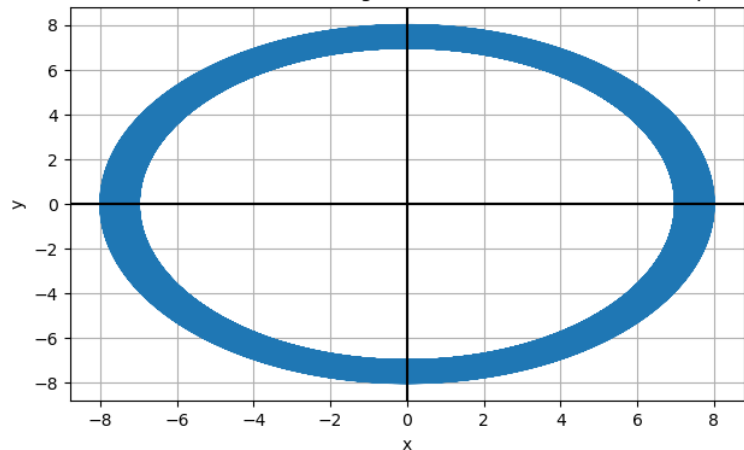


Figure 17

Radial Motion of a Particle Orbiting a Black Hole in Schwarzschild Spacetime

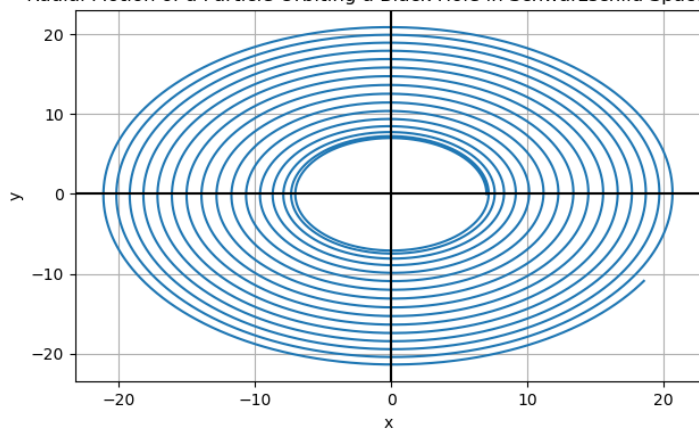


Figure 18

circle as seen in figure 17 (for one that might not do this will have the angular rotation of the ellipse repeat after 8 periods, making a flower shape with 8 petals, this may be possible but I have yet to find it in my search and given how the thickness changes with α I think it may be impossible, I just have no way to prove it). When plotting figure 18, it does not match up with what Schutz says should be happening, since it appears to be a spiral instead of rotating ellipses, I'm not sure what has caused this but I am able to identify that it does not look right compared to the figures in the book and my knowledge based on reading chapter 11.

Section (b): Biography of (scientist involved in general relativity)

Andrea Ghez

Black holes are astronomical bodies that are so dense that even light cannot escape, thought about by scientists for hundreds of years like John Michell and Pierre-Simon Laplace, both of which proposed stars like modern models of supermassive black holes, which are the largest type of black hole, long before Einstein came up with his theory of general relativity [1]. One of which was proved to be at the centre of our galaxy, the Milky Way, by a world-renowned physicist Andrea Ghez (b. 1965). It was first postulated by Donald Lynden-Bell and Martin Rees in 1971[2], but they were unable to prove its existence unlike Ghez, who provided evidence for its existence by tracking orbits of individual stars near the centre of the galaxy with precision like no other [3]. It is said that her work in general relativity on the orbits of stars at the centre of the Milky Way has opened a new approach to studying black holes and she even won a Nobel prize in physics for her work in this field [3], being the fourth woman ever to receive this award out of only five to date.

Ghez, the Physicist

Born in New York City, Ghez was raised by her parents Susanne and Gilbert Ghez, of which Gibert obtained a PhD at Columbia University and accepted a position at the University of Chicago in 1969 when Andrea was young meaning that, since she was a child of a member of staff, Andrea was able to attend the Laboratory School, already starting a life of academics from a young age. She was inspired by the Apollo program and wanted to become the first female astronaut, even being purchased a telescope by her mother, fuelling her interest in the stars even more and pushing her in the direction of her eventual career in astrophysics, testing Einsteins theory of general relativity.

Later in life she began college, majoring in mathematics initially but changing to physics later, receiving her BS in physics from the Massachusetts Institute of Technology in 1987 and went on to start a PhD at the California Institute of Technology, earning it in 1992.

For her PhD, her thesis was titled 'The Multiplicity of T Tauri Stars in the Star Forming Regions Taurus-Auriga and Ophiuchus-Scorpius: A 2.2 Micron Speckle Imaging Survey' [4] and involved using infrared speckle imaging to see how often these stars occur in binary or multiple systems and although her PhD was not directly related to general relativity, it allowed her to develop skills she used to test the theory of relativity later in her career. This paper had many citations from it however despite it being her first scientific paper, being her 8th most cited paper to date with 206 citations.

After earning her PhD with great success, Ghez joined the faculty at the University of California (UCLA) in 1994 and ended up becoming leader of the Galactic Centre Group, a research team that is focused on studying the centre of the Milky Way. In her time at UCLA, she wrote and published many famous and well-known paper that got thousands of citations related to testing Einsteins general relativity, her most cited at present being her paper 'Measuring Distance and Properties of the Milky Way's Central Supermassive Black Hole with Stellar Orbits' with 1,158 citations [5]. This paper provides a large foundation that makes testing general relativity in strong gravity possible by getting extremely accurate knowledge of the black hole's mass and distance from Earth needed to compare observed orbits with general relativity predictions.

After this, Ghez continued to publish papers related to general relativity, the most direct one in 2017 being 'Testing General Relativity with Stellar Orbits around the Supermassive Black Hole in Our Galactic Centre' [6], in this paper, Ghez and the other scientists that worked on the paper used decades of observations on stars orbiting close to the Milky Way's central black hole to act like natural test particles in a strong gravitational field, so by tracking their positions and velocities, she could determine if it matched with what Einstein's theory of general relativity predicts. The result of this test closely matched with the predictions, meaning that Einstein's equations hold in that environment which was very big at the time since it was one of the first high precision tests under extreme conditions and provided evidence for the equations validity. This also led on to another paper that Ghez wrote with other scientists about detection of relativistic redshift.

This paper was 'Relativistic Redshift of the Star S0-2 Orbiting the Galactic Centre Supermassive Black Hole' [7] released in 2019. The paper showed precise measurements of the light coming off of star S0-2 as it made its closest

approach to the supermassive black hole at the centre of the Milky Way and the results were exactly what Einstein had predicted, that a star getting close enough to the centre of a strong gravitational field would have its light stretched by the gravity and speed at which the star is moving, making the light show redder. This matched with relativistic redshift and was amazing further evidence for general relativity, even being shown to be exclusive to when the star was close enough to the supermassive black hole.

The Nobel Prize

In 2020, Ghez shared half of the Nobel prize in physics with Reinhard Genzel for their work in “the discovery of a supermassive compact object at the centre of our galaxy” [8]. This was a massive accomplishment for her and used the platform to highlight the importance of women and other underrepresented groups in science. She was quoted saying "I'm thrilled to receive the prize, and I take very seriously the responsibility of being the fourth woman to win the Nobel prize [in physics]" and became an icon for women in STEM everywhere.

Her work that led to her receiving this prize was from her work tracking stars orbits in the centre of our galaxy, as shown in papers mentioned earlier, and using this to prove that the centre of the Milky Way contains a supermassive black hole. Her tireless work from decades of incremental progress on mapping orbits, velocities and masses of stars thousands of lightyears away paid off and she is known across the world for these methods and discoveries that helped test Einstein's general relativity in ways never thought possible before her work and the prize recognises her for far more than a single discovery.

References

- [1] C. Montgomery, W. Orchiston, & I. Whittingham, (2009). *Mitchell, Laplace and the origin of the black hole concept*. **Journal of Astronomical History and Heritage**, 12, 90–96. <https://doi.org/10.3724/SP.J.1440-2807.2009.02.01>
- [2] D. Lynden-Bell, & M. Rees, (1971). *On quasars, dust and the galactic centre*. Monthly Notices of the Royal Astronomical Society, 152, 461. <https://doi.org/10.1093/mnras/152.4.461>
- [3] "[Andrea Ghez](https://www.nobelprize.org/andrea-ghez)". NobelPrize.org.

- [4] A. Ghez, G. Neugebauer, & K. Matthews, (1993). The multiplicity of T Tauri stars in the star-forming regions Taurus-Auriga and Ophiuchus-Scorpius: A 2.2 μm speckle imaging survey. *Astronomical Journal*, **106**, 2005–2023. <https://doi.org/10.1086/116782>
- [5] A. Ghez, et al. (2008). Measuring distance and properties of the Milky Way's central supermassive black hole with stellar orbits. *The Astrophysical Journal*, 689, 1044–1062. <https://doi.org/10.1086/592738>
- [6] A. Ghez, et al. (2017). Testing General Relativity with stellar orbits around the supermassive black hole in our Galactic center. *Physical Review Letters*, 118(21), 211101. <https://doi.org/10.1103/PhysRevLett.118.211101>
- [7] A. Ghez, et al. (2019). Relativistic redshift of the star S0-2 orbiting the Galactic center supermassive black hole. *Science*, 365(6454), 664–668. <https://doi.org/10.1126/science.aav8137>
- [8] [Press release: The Nobel Prize in Physics 2020 - NobelPrize.org](#)